Customer Churn – Data Quality & Exploratory Analysis Report

Dataset size: 15 customer records and 11 fields. Overall churn rate: 46.7%.

1. Data Quality Assessment

Check	Result
Rows / columns	15 / 11
Missing values	0
Duplicate rows	0
Invalid numeric ranges	None found
TotalCharges consistency	Passed
IQR outliers	None found

2. Cleaning Decisions

• Trimmed leading/trailing whitespace in text fields.

• Removed exact duplicate rows; none were present.

• Checked missing values; none were present.

• Checked numeric ranges for age, tenure, charges, and support tickets; no invalid values were found.

• Checked TotalCharges ≈ MonthlyCharges × TenureMonths; all records passed within ■0.01 rounding tolerance.

• No IQR outliers were found in the numeric columns, so no observations were removed as outliers.

3. Summary Statistics

Variable	Mean	Median	Min	Max
Age	39.07	38.00	23.00	60.00
TenureMonths	18.80	15.00	2.00	48.00
MonthlyCharges	84.66	79.99	49.99	149.99
TotalCharges	2088.48	1099.78	99.98	7199.52
SupportTickets	2.53	2.00	0.00	6.00

4. Initial EDA Findings

• Month-to-month customers have a 100.0% churn rate in this sample, while one-year and two-year contract customers show 0% churn.

• Basic subscription customers show a 71.4% churn rate; Enterprise customers show 0% in this sample.

• Customers who churned have a lower average tenure (7.0 months) than customers who did not (29.1 months).

• Churned customers average 4.29 support tickets versus 1.00 for non-churned customers.

• Because the dataset contains only 15 records, these are descriptive patterns rather than statistically reliable population-level conclusions.

5. Validation Conclusion

The cleaned dataset is analysis-ready for the scope of this task. No missing values, duplicate rows, invalid ranges, charge inconsistencies, or IQR-based outliers were identified. The original values were therefore retained after validation.